

DATA TRANSFORMATION

Definition : Cleaning and filtering the data before storing it in Power BI's internal model, so it can be used for modeling and further steps.

Tool Used: Power Query Editor.

What is a Power Query?

- Power Query is Power BI's data preparation engine. Performs ETL Process
- **Purpose:** Connect to a data source, reshape the data, and load it into Power BI's internal model — **without touching the original source**.
- **M Language:** Every transformation is recorded as a **step** in a script called **M** (the underlying language). Power Query generates M automatically as you click.

Step-by-Step: Loading Your First Dataset

1. **Open Power BI Desktop** (Download free from microsoft.com/powerbi).
2. **Get Data:** Click Home → Get Data → Text/CSV (Starting with CSV, the simplest format).
3. **Load sample data:** Use this free dataset: [Superstore Sales CSV](#). (Classic retail data: orders, products, regions, profit).
4. **Enter Power Query Editor:** After selecting the file, click **Transform Data** (not Load). This opens the Power Query Editor window.

Understanding Power Query Editor Interface

Panel	What it does
Left sidebar	List of queries (tables)
Center	Preview of data
Right sidebar	Applied Steps — every transformation recorded
Top ribbon	Transformation tools

The Three Basic Transformations

1. Choose the Columns:

Purpose: Select which columns to keep based on requirements.

Method 1: Choose Columns (Keep/Deselect)

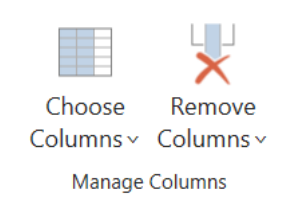
Step 1: Power Query Editor > Home >

Step 2: Manage columns > Choose Columns

Step 3: Check the box next to columns to keep (or deselect unwanted ones)

Step 4: Click OK

Example: In Superstore, removed **Row ID** from the *Orders* table.



Method 2: Remove Columns (Remove Selected/Other)

Step 1: Select the columns to remove (Ctrl+select for multiple)

Step 2: Choose **Remove Columns** (removes selected ones) **OR Remove other Columns** (removes all *except* selected ones)

Example: Removed **Postal Code** (using Remove columns > Remove columns).

2. Rename the Columns:

Purpose: Rename columns that have large names or are hard to process/understand.

Method 1: Right-Click

Step 1: Select the column

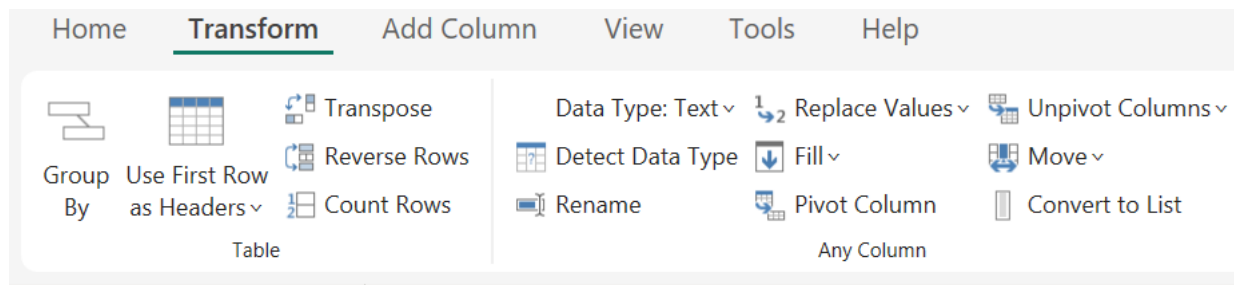
Step 2: Right-click on the column header > Rename

Step 3: Type the new name then press ENTER

Method 2: Transform Tab

Step 1: Select column

Step 2: Transform > Any column: Rename



3. Change the type (Data Type):

Purpose: Power Query often defaults all columns to Text, which can be dangerous for insights. Need to change the data type.

Method 1: Right-Click

Step 1: Select column > Right click

Step 2: Change Type > Choose the data type of the column > press ENTER

Method 2: All columns at once

Step 1: Select All (Ctrl+A)

Step 2: Transform > Any columns > Detect DataType

Example: In the *Peoples Table*, the data type of both columns was "Any" and was changed to **Text**.

SOME OTHER TRANSFORMATIONS

1. Column Header:

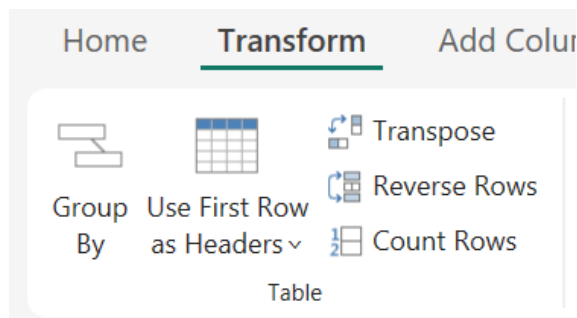
Purpose: The table must have proper column headers (especially if headers are in the first row) to get proper insights.

Step 1: Select the columns

Step 2: Transform > Use First row as Headers

- (Options are: Use First Row as Header OR Use Header as First row)

Step 3: Select **Use First Row as Header**



2. Splitting the column:

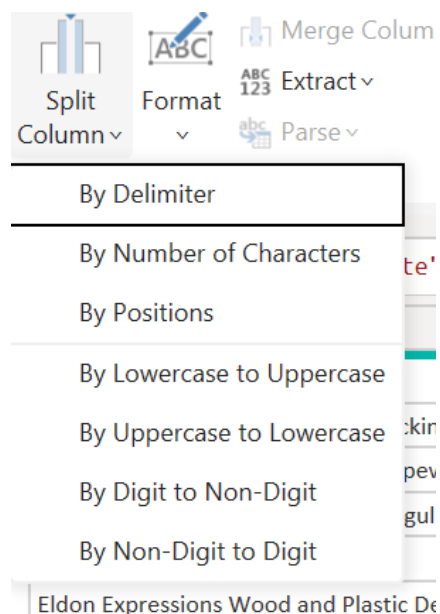
Purpose: Separate data within a single column that needs to be isolated for searching, filtering, or deeper analysis.

Step 1: Select the column

Step 2: Home > Transform > Split Column

Step 3: Select the preferred split method > click OK

Example: Split **Customer Name** into two distinct columns using the **By Delimiter** option.



3. Remove the white spaces in the start and end:

Step 1: Select the column

Step 2: Transform > Format > Trim

